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Dense diversification or sparse allocation? Evidence from high-dimensional minimum variance portfolios

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Abstract

This study investigates whether high-dimensional minimum variance portfolios benefit more from sparse allocation or from dense diversification. Since the minimum variance portfolio weight depends on the covariance matrix through $\Sigma^{-1}\mathbf{1}$, estimation of weight can be performed at different levels: estimating the covariance matrix, estimating the precision matrix, or directly estimating this portfolio-relevant vector. We compare covariance-based plug-in estimators, including POET, nonlinear shrinkage, and a James–Stein type covariance estimator, with sparsity-inducing methods such as graphical lasso and a Dantzig-type estimator. Using daily return data from the KOSPI and S&P 500 markets, we evaluate out-of-sample portfolio risk by the empirical standard deviation of rolling-window realized returns. The results show that dense covariance-based estimators generally outperform estimators that impose sparsity on the precision matrix or on the portfolio-relevant vector $\Sigma^{-1}\mathbf{1}$, suggesting that such sparsity can lead to overly concentrated portfolios and weaken the benefit of broad diversification. Among the dense methods, estimators that retain the leading common component of asset returns, particularly POET and the proposed James–Stein based estimator, provide stable portfolio performance. Overall, the findings indicate that, in the high-dimensional settings considered, preserving dense diversification through structured covariance estimation is more effective than imposing sparsity directly on portfolio allocation.

Keywords: portfolio risk, high-dimensional covariance estimation, James–Stein estimator

1. Introduction

A minimum variance portfolio is designed to find portfolio weights w that minimize portfolio variance under the fully invested condition $w^\top \mathbf{1} = 1$. Given the covariance matrix $\Sigma \in \mathbb{R}^{p\times p}$ of returns on p assets, the population minimum variance portfolio weight is defined as

$$w^* = \underset{w^\top \mathbf{1}=1,\, w\in\mathbb{R}^p}{\operatorname{argmin}} \; w^\top \Sigma w = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top \Sigma^{-1}\mathbf{1}}.$$

This expression shows that the minimum variance portfolio depends directly on the inverse covariance matrix. In high-dimensional settings, where the number of assets p is comparable to or larger than the sample size n, the sample covariance matrix is difficult to estimate accurately and its inverse can be highly unstable (Ledoit and Wolf, 2004; Fan *et al.*, 2013). Since estimation error in the covariance matrix can be amplified through portfolio optimization, the choice of covariance or precision matrix

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estimator can substantially affect the estimated weights and the resulting portfolio risk (Jagannathan and Ma, 2003).

Since the optimal weight vector depends on the covariance matrix only through the quantity $\Sigma^{-1}\mathbf{1}$, this portfolio-relevant quantity can be approached in three different ways: by estimating the covariance matrix, by estimating the precision matrix, or by directly estimating $\Sigma^{-1}\mathbf{1}$. In this paper, we compare five estimators that reflect these three estimation approaches. The first approach consists of covariance plug-in methods: Principal Orthogonal complEment Thresholding (POET) (Fan *et al.*, 2013), nonlinear shrinkage (NLS) (Ledoit and Wolf, 2020), and a James–Stein (JS) type covariance estimator (Shkolnik *et al.*, 2025; Yoon and Jung, 2025). These estimators regularize the sample covariance matrix through different structural assumptions, including approximate factor structure, eigenvalue shrinkage, and eigenvector shrinkage toward a given reference direction. Because they typically produce dense covariance estimates, the resulting minimum variance portfolios are also generally dense.

A key component of our study is a James–Stein type covariance estimator based on eigenvector shrinkage. The James–Stein method for eigenvectors framework shrinks sample eigenvectors toward reference directions or a target subspace, thereby reducing the estimation error of leading principal components in high-dimensional settings (Yoon and Jung, 2025; Goldberg *et al.*, 2025). For example, Yoon and Jung (2025) propose a reference-guided estimator of the principal component subspace, whereas Goldberg *et al.* (2025) use a generalized James–Stein method for eigenvectors approach to construct covariance estimators tailored to portfolio optimization. Building on this line of work, Shkolnik *et al.* (2025) applies eigenvector shrinkage to covariance matrix estimation for minimum variance portfolio construction.

Under a factor model assumption, the covariance matrix can be decomposed into a low-rank common component and an idiosyncratic covariance component (Chamberlain and Rothschild, 1983). In the JS type covariance estimator considered by Shkolnik *et al.* (2025), the residual covariance component is regularized by retaining its diagonal part. In this paper, we modify this treatment of the residual covariance matrix by adopting the approximate factor model perspective underlying POET (Fan *et al.*, 2013). Instead of diagonalizing the residual component, we apply thresholding to the residual covariance matrix, allowing for a sparse idiosyncratic covariance structure. This yields a covariance estimator that combines eigenvector shrinkage for the low-rank component with residual thresholding for the idiosyncratic component. As another plug-in method, the nonlinear shrinkage method (Ledoit and Wolf, 2020) constructs a covariance estimator by preserving the sample eigenvectors of the sample covariance matrix and applying a nonlinear shrinkage transformation to the sample eigenvalues.

The second approach estimates the precision matrix directly. We use the graphical lasso (Friedman *et al.*, 2008), which imposes an ℓ_1 penalty on the inverse covariance matrix and is suited to settings where the precision matrix is sparse. The third approach bypasses both covariance and precision matrix estimation by directly targeting the vector $\Sigma^{-1}\mathbf{1}$. For this purpose, we adopt a Dantzig-type estimator (Candès and Tao, 2007), which imposes regularization on $\Sigma^{-1}\mathbf{1}$ itself.

The five methods considered in this paper differ not only in the object they estimate, but also in whether they produce dense portfolio weights or induce sparsity in the resulting allocation. Throughout this paper, the terms *dense* and *sparse* refer to the structure of the estimated portfolio weight vector. A dense allocation is one in which most assets receive nonzero weights, so that the portfolio remains broadly diversified across the assets. A sparse allocation is one in which many weights are zero or nearly zero, so that the portfolio is effectively concentrated on a smaller subset of assets. Thus, the terms dense diversification and sparse allocation describe the allocation pattern induced by each estimator, rather than only the sparsity structure of the covariance matrix, precision matrix, or

portfolio-relevant vector $\Sigma^{-1}\mathbf{1}$. This distinction leads to the central practical question of this paper: in high-dimensional minimum variance portfolio construction, is it more effective to impose sparsity on the allocation, or to preserve broad diversification through structured covariance estimation?

To answer this question, we evaluate the performance of each method using daily return data from the KOSPI and S&P 500 markets. The datasets are divided according to whether the COVID-19 period is included, since the strength of the leading market component differs substantially across the two sample designs. For each method, portfolio weights are constructed through a rolling-window procedure, and performance is measured by the empirical standard deviation of the out-of-sample realized returns. This empirical design allows us to compare not only which estimator produces lower realized risk, but also which type of regularization leads to more stable portfolio construction in practice.

The empirical results provide two main insights. First, dense estimators based on covariance matrix estimation tend to outperform estimators that induce sparsity through precision matrix estimation or direct estimation of $\Sigma^{-1}\mathbf{1}$. In particular, graphical lasso and the Dantzig-type estimator produce relatively large realized standard deviations compared with the covariance-based methods. This suggests that, in the minimum variance portfolio problem considered here, directly imposing sparsity on the precision matrix or on the portfolio-relevant vector $\Sigma^{-1}\mathbf{1}$ may lead to overly concentrated allocations and reduce the benefit of diversification. From a practical perspective, reducing estimation error through sparsity does not necessarily translate into lower portfolio risk if the resulting portfolio loses broad exposure across assets.

Second, among the dense covariance-based methods, estimators that explicitly reflect the leading common component of asset returns perform particularly well. POET is the most robust method overall, while the proposed JS-based estimator remains competitive and performs especially well in some KOSPI settings. This finding supports the usefulness of a low-rank plus regularized residual covariance structure for portfolio construction. In particular, methods such as POET and the proposed JS-based estimator stabilize the covariance estimate while preserving the dominant direction of market co-movement. Overall, our analysis suggests that high-dimensional minimum variance portfolios benefit more from structured covariance regularization that preserves dense diversification than from sparsity imposed directly on the allocation.

2. Weight estimators

This section defines the five weight estimators used in the empirical analysis. We classify them according to the object being regularized and the type of allocation they tend to produce. POET, the proposed JS-based estimator, and NLS are covariance-based plug-in methods that generally lead to dense portfolio weights. Graphical lasso regularizes the precision matrix, while the Dantzig-type estimator directly regularizes the portfolio-relevant vector $\Sigma^{-1}1$. Both methods may induce sparse portfolio allocations.

2.1. Dense estimators

2.1.1. POET estimator

Principal Orthogonal complEment Thresholding (POET) (Fan *et al.*, 2013) is a covariance estimator designed for high-dimensional covariance matrices with an approximate factor model. Denote a data matrix as $Y \in \mathbb{R}^{p\times T}$ and its sample covariance matrix $\widehat{\Sigma}_{\text{sam}} = \frac{1}{T}YY^{\top}$. Let

$$\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \cdots \geq \hat{\lambda}_p$$

be its ordered eigenvalues with corresponding eigenvectors $\{\hat{\xi}_i\}_{i=1}^p$. For a given number of factors K, the sample covariance matrix can be decomposed into the leading K principal components and the remaining residual component as

$$\widehat{\Sigma}_{\text{sam}} = \sum_{i=1}^{K} \hat{\lambda}_i \hat{\xi}_i \hat{\xi}_i^\top + \widehat{R}_K,$$

where

$$\widehat{R}_K = \sum_{i=K+1}^{p} \hat{\lambda}_i \hat{\xi}_i \hat{\xi}_i^\top = (\hat{r}_{ij})_{p\times p}.$$

POET is motivated by the assumption that the population covariance matrix can be decomposed into a low-rank common component and a sparse idiosyncratic component.

The residual component $\widehat{R}_K$ is further regularized by adaptive thresholding (Cai and Liu, 2011). Adaptive thresholding refers to the use of entry-specific thresholding levels τ_{ij}, which account for the variability of each residual covariance entry. Given these thresholds, we use the soft-thresholding rule to shrink the off-diagonal entries. Let

$$\widehat{R}_K^{\mathcal{T}} = \left(\hat{r}_{ij}^{\mathcal{T}}\right)_{p\times p}$$

denote the thresholded version of $\widehat{R}_K$. Its entries are defined as

$$\hat{r}_{ij}^{\mathcal{T}} = \begin{cases} \hat{r}_{ii}, & i = j, \\ s_{ij}(\hat{r}_{ij}), & i \neq j, \end{cases}$$

where $s_{ij}(\cdot)$ is defined as follows:

$$s_{ij}(x) = \text{sign}(x)(|x| - \tau_{ij})_+, \qquad (x)_+ = \max\{x, 0\}.$$

To define the adaptive thresholding parameter τ_{ij}, following Cai and Liu (2011), we consider an approximate factor model

$$y_{it} = b_i^\top f_t + u_{it}, \quad i = 1, \ldots, p, \ t = 1, \ldots, T,$$

where y_{it} is the observed return of asset i at time t, b_i is a vector of factor loadings, f_t is a $K \times 1$ vector of common factors, and u_{it} is the idiosyncratic error term. In matrix form, this can be written as

$$Y = BF^\top + E,$$

where $Y \in \mathbb{R}^{p\times T}$, $B \in \mathbb{R}^{p\times K}$, $F \in \mathbb{R}^{T\times K}$, and $E = (u_{it})_{i\leq p, t\leq T}$. The i-th row of B is $b_i^\top$, and the t-th row of F is $f_t^\top$. According to Fan *et al.* (2013), the common factors and factor loadings are estimated by principal components. Let

$$Y = \widehat{\Xi}\widehat{D}\widehat{V}^\top$$

be the singular value decomposition of Y, where $\widehat{\Xi} = [\hat{\xi}_1, \ldots, \hat{\xi}_p]$ and the columns of $\widehat{V}$ are the right singular vectors. Specifically, we use

$$\widehat{F} = \sqrt{T}\,\widehat{V}_{(:,1:K)}, \quad \widehat{B} = \frac{1}{T} Y\widehat{F}.$$

The residuals are estimated by

$$\hat{u}_{it} = y_{it} - \hat{b}_i^\top \hat{f}_t,$$

and the entrywise variance of the residual covariance $\widehat{R}_K$ are estimated as

$$\hat{\theta}_{ij} = \frac{1}{T}\sum_{t=1}^{T}\left(\hat{u}_{it}\hat{u}_{jt} - \hat{r}_{ij}\right)^2.$$

Adaptive thresholding parameter is given by

$$\tau_{ij} = C\sqrt{\hat{\theta}_{ij}}\, w_T, \tag{2.1}$$

where $C > 0$ is a tuning constant and

$$w_T = \frac{1}{\sqrt{p}} + \sqrt{\frac{\log p}{T}}.$$

The resulting POET covariance estimator is then defined as

$$\widehat{\Sigma}_{\text{POET}} = \sum_{i=1}^{K}\hat{\lambda}_i\hat{\xi}_i\hat{\xi}_i^\top + \widehat{R}_K^{\mathcal{T}},$$

where

$$\widehat{R}_K^{\mathcal{T}} = (\hat{r}_{ij}^{\mathcal{T}})_{p\times p}, \quad \hat{r}_{ij}^{\mathcal{T}} = \begin{cases} \hat{r}_{ii}, & i = j, \\ \text{sign}(\hat{r}_{ij})(|\hat{r}_{ij}| - \tau_{ij})_+, & i \neq j. \end{cases}$$

Since entrywise thresholding does not necessarily preserve positive definiteness, we choose the thresholding constant C so that $\widehat{R}_K^{\mathcal{T}}$ is positive definite. For any $C > 0$, let $\widehat{R}_K^{\mathcal{T}}(C)$ denote the residual covariance estimator obtained with thresholding constant C appearing in (2.1). As C increases, the off-diagonal entries of $\widehat{R}_K^{\mathcal{T}}(C)$ shrink to zero, so the estimator approaches a diagonal matrix. Thus, provided that the diagonal entries of $\widehat{R}_K^{\mathcal{T}}(C)$ remain strictly positive, there exists a sufficiently large C such that minimum eigenvalue of $\widehat{R}_K^{\mathcal{T}}(C)$ is strictly positive. We define

$$C_{\min} = \inf\left\{C > 0 : \lambda_{\min}\left(\widehat{R}_K^{\mathcal{T}}(M)\right) > 0 \quad \text{for all } M > C\right\}.$$

In the implementation, we choose $C = C_{\min} + \epsilon$. In this study, we set $\epsilon = 0.1$.

Then the POET estimator can be written as

$$\widehat{\Sigma}_{\text{POET}} = \widehat{R}_K^{\mathcal{T}} + \widehat{B}\widehat{B}^\top,$$

which, by construction, is invertible. By the Woodbury identity, its inverse is given by

$$\widehat{\Sigma}_{\text{POET}}^{-1} = (\widehat{R}_K^{\mathcal{T}})^{-1} - (\widehat{R}_K^{\mathcal{T}})^{-1}\widehat{B}\left(I_K + \widehat{B}^\top(\widehat{R}_K^{\mathcal{T}})^{-1}\widehat{B}\right)^{-1}\widehat{B}^\top(\widehat{R}_K^{\mathcal{T}})^{-1}.$$

Finally, the minimum variance portfolio weight estimator is given by

$$\hat{w}_{\text{POET}} = \frac{\widehat{\Sigma}_{\text{POET}}^{-1}\mathbf{1}}{\mathbf{1}^\top\widehat{\Sigma}_{\text{POET}}^{-1}\mathbf{1}}.$$

2.1.2. James Stein estimator

In the high-dimensional, low-sample size (HDLSS) regime where $p \to \infty$ and n is fixed, the leading sample eigenvectors corresponding to spiked eigenvalues are known to be inconsistent (Jung *et al.*, 2012). To mitigate this inconsistency, James Stein (JS) method aims to estimate the true Principal Component (PC) subspace, $\mathcal{U}_K = \text{span}(\xi_1, \ldots, \xi_K)$, by combining the sample eigenvectors with additional vectors that contain the prior information of the true PC subspace. These additional vectors are referred to as reference directions and are denoted by $v_1, \ldots, v_r$.

Define the subspace $\mathcal{S} = \text{span}(\hat{\xi}_1, \ldots, \hat{\xi}_K, v_1, \ldots, v_r)$. Then our goal is to find a subspace within $\mathcal{S}$ that is closest to $\mathcal{U}_K$. Then it naturally leads to think of the orthogonal projection of $\mathcal{U}_K$ onto $\mathcal{S}$, i.e, $P_{\mathcal{S}}\mathcal{U}_K$. Yoon and Jung (2025) and Shkolnik *et al.* (2025) introduce negatively ridged discriminant vectors $d_1, \ldots, d_r$, defined as follows:

$$d_i := -\tilde{\lambda}(S_K - \tilde{\lambda} I_p)^{-1} v_i, \quad \text{where} \quad \tilde{\lambda} = \sum_{i=K+1}^{T-1} \hat{\lambda}_i/(T-K-1),$$

where $S_K = \sum_{i=1}^{K} \hat{\lambda}_i \hat{\xi}_i \hat{\xi}_i^\top$. Then, the asymptotic orthogonality between $\mathcal{D}_r = \text{span}(d_1, \ldots, d_r)$ and $\mathcal{U}_K$ leads to estimate $P_{\mathcal{S}}\mathcal{U}_K$ as $\mathcal{S} \setminus \mathcal{D}_r$. The estimated PC subspace is then calculated as follows:

$$\widehat{\mathcal{U}}_K = \text{span}((S_K - \tilde{\lambda} I_p)(I_p - P_{\mathcal{V}_r})\widehat{U}_K),$$

where $\mathcal{V}_r = \text{span}(v_1, \ldots, v_r)$, $\widehat{U}_K = [\hat{\xi}_1, \ldots, \hat{\xi}_K]$.

We then project the data onto $\widehat{\mathcal{U}}_K$ and define

$$\widetilde{Y} = P_{\widehat{\mathcal{U}}_K} Y.$$

Let $\widetilde{Y} = \widetilde{U}\widetilde{D}\widetilde{V}^\top$ be its singular value decomposition, where the columns of $\widetilde{V}$ are the right singular vectors. Define the estimated factor matrix and loading matrix by

$$\widetilde{F} = \sqrt{T}\widetilde{V}_{(:,1:K)}, \quad \widetilde{B} = \frac{1}{T}\widetilde{Y}\widetilde{F}.$$

The low rank component of Σ is estimated by $\widehat{L} = \widetilde{B}\widetilde{B}^\top = \frac{1}{T}\widetilde{Y}\widetilde{Y}^\top = P_{\widehat{\mathcal{U}}_K} S P_{\widehat{\mathcal{U}}_K}$.

We estimate the residual component of Σ using the POET thresholding procedure. We first define the residual data matrix as

$$\widetilde{E} = Y - \widetilde{B}\widetilde{F}^\top = Y - \widetilde{Y} = (I - P_{\widehat{\mathcal{U}}_K})Y = (\tilde{u}_{it})_{1 \le i \le p, 1 \le t \le T}.$$

Then, the residual covariance matrix is estimated as

$$\widetilde{\Sigma}_E = \frac{1}{T}\widetilde{E}\widetilde{E}^\top = (\tilde{\sigma}_{ij})_{1 \le i,j \le p}, \quad \tilde{\sigma}_{ij} = \frac{1}{T}\sum_{t=1}^{T} \tilde{u}_{it}\tilde{u}_{jt}$$

Following the adaptive thresholding idea of POET, we threshold the off-diagonal entries of $\widetilde{\Sigma}_E$. For each i, j, define

$$\tilde{\theta}_{ij} = \frac{1}{T}\sum_{t=1}^{T} (\tilde{u}_{it}\tilde{u}_{jt} - \tilde{\sigma}_{ij})^2,$$

so that $\tilde{\theta}_{ij}^{1/2}$ is the sample standard deviation of $(\tilde{u}_{i1}\tilde{u}_{j1}, \ldots, \tilde{u}_{in}\tilde{u}_{jn})$. Let $w_T = \frac{1}{\sqrt{p}} + \sqrt{\frac{\log p}{T}}$, $\gamma_{ij} = C' w_T \tilde{\theta}_{ij}^{1/2}$, where $C' > 0$ is a tuning constant. $\widetilde{R}_K^{\mathcal{T}}$ denote the thresholded version of $\widetilde{\Sigma}_E$, then,

$$\widetilde{\mathcal{R}}_K^{\mathcal{T}} = (\tilde{\sigma}_{ij}^{\mathcal{T}})_{1\le i,j\le p}, \quad \tilde{\sigma}_{ij}^{\mathcal{T}} = \begin{cases} \tilde{\sigma}_{ii}, & i = j, \\ \text{sign}(\tilde{\sigma}_{ij})(|\tilde{\sigma}_{ij}| - \gamma_{ij})_+, & i \neq j, \end{cases}$$

where $(.)_+ = \max(., 0)$. The final JS-thresholding covariance estimator is $\widehat{\Sigma}_{\text{JS}} = \widehat{L} + \widetilde{\mathcal{R}}_K^{\mathcal{T}}$. In this study, we choose $C' = C'_{\min} + \epsilon$, $\epsilon = 0.1$, where $C'_{\min}$ is the smallest value such that the resulting residual covariance estimator $\widetilde{\mathcal{R}}_K^{\mathcal{T}}$ is positive definite.

For the inverse matrix, we use the Woodbury-type formula. Since

$$\widehat{\Sigma}_{\text{JS}} = \widetilde{\mathcal{R}}_K^{\mathcal{T}} + \widetilde{B}\widetilde{B}^\top,$$

we have

$$\widehat{\Sigma}_{\text{JS}}^{-1} = (\widetilde{\mathcal{R}}_K^{\mathcal{T}})^{-1} - (\widetilde{\mathcal{R}}_K^{\mathcal{T}})^{-1}\widetilde{B}\left(I_K + \widetilde{B}^\top(\widetilde{\mathcal{R}}_K^{\mathcal{T}})^{-1}\widetilde{B}\right)^{-1}\widetilde{B}^\top(\widetilde{\mathcal{R}}_K^{\mathcal{T}})^{-1}.$$

The corresponding minimum variance portfolio weight estimator is

$$\hat{w}_{\text{JS}} = \frac{\widehat{\Sigma}_{\text{JS}}^{-1}\mathbf{1}}{\mathbf{1}^\top\widehat{\Sigma}_{\text{JS}}^{-1}\mathbf{1}}.$$

The covariance estimator considered in Shkolnik *et al.* (2025) is a special case of the JS estimator. This estimator is given by taking a sufficiently large thresholding parameter C' so that $\widetilde{\mathcal{R}}_K^{\mathcal{T}}$ becomes a diagonal matrix, and the resulting weight estimator is called JS-diag estimator.

2.1.3. Nonlinear shrinkage estimator

Nonlinear shrinkage (NLS) estimator is a covariance matrix estimator designed for large-dimensional settings in which the dimension p is comparable to the sample size n. In classical low-dimensional asymptotics, the sample covariance matrix is a consistent estimator of the population covariance matrix. However, when $p/n \to c \in (0, \infty)$, the sample eigenvalues are no longer reliable estimates of the population eigenvalues. Random matrix theory shows that the empirical distribution of the sample eigenvalues is systematically distorted relative to the population spectral distribution (Marčenko and Pastur, 1967; Silverstein and Bai, 1995; Ledoit and Wolf, 2012). This distortion is characterized by the Marčenko–Pastur equation, which links the limiting spectral distribution of the sample covariance matrix to that of the population covariance matrix.

The main idea of nonlinear shrinkage is therefore to keep the sample eigenvectors $\hat{\xi}_i$, but to replace the sample eigenvalues $\hat{\lambda}_i$ with adjusted values. Unlike linear shrinkage (Ledoit and Wolf, 2004), which shrinks all eigenvalues toward a common target with the same intensity, nonlinear shrinkage applies a different transformation to each eigenvalue. This allows various size of eigenvalues to be adjusted differently according to their locations in the spectrum (See Figure 2 in Ledoit and Wolf (2020)).

Ledoit and Wolf (2012) derive an oracle nonlinear shrinkage function based on the limiting spectral distribution. In the analytical nonlinear shrinkage formulation of Ledoit and Wolf (2020), the oracle shrinkage function can be written as

$$d^\circ(x) = \frac{x}{[\pi c x f(x)]^2 + \left[1 - c - \pi c x \mathcal{H}_f(x)\right]^2},$$

where $c = \lim_{n\to\infty} p/n$, f is the limiting spectral density of the sample eigenvalues, and $\mathcal{H}_f$ denotes the Hilbert transform of f. This expression shows that the amount of shrinkage depends nonlinearly on the position x of each eigenvalue in the sample spectrum.

In practice, the population spectral distribution and the oracle shrinkage function are unknown. Ledoit and Wolf (2020) estimate the density f and its Hilbert transform $\mathcal{H}_f$ directly from the observed sample eigenvalues using kernel methods. In implementation, we use kernel smoothing with the Epanechnikov kernel and the specific formula is provided in the Appendix A. Let $\hat{f}$ and $\widehat{\mathcal{H}_f}$ denote the kernel estimates of f and $\mathcal{H}_f$, respectively. Then the estimated nonlinear shrinkage function is given by

$$\hat{d}(x) = \frac{x}{\left[\pi \frac{p}{n} x \hat{f}(x)\right]^2 + \left[1 - \frac{p}{n} - \pi \frac{p}{n} x \widehat{\mathcal{H}_f}(x)\right]^2}.$$

Applying this function to each sample eigenvalue yields the nonlinearly shrunk eigenvalues

$$\hat{\lambda}_i^{\mathrm{NLS}} = \hat{d}(\hat{\lambda}_i), \quad i = 1, \ldots, p.$$

The nonlinear shrinkage covariance estimator is then constructed as

$$\widehat{\Sigma}_{\mathrm{NLS}} = \sum_{i=1}^{p} \hat{\lambda}_i^{\mathrm{NLS}} \hat{\xi}_i \hat{\xi}_i^\top.$$

Thus, NLS regularizes the covariance matrix through the eigenvalue spectrum while leaving the sample eigenvectors unchanged. The minimum variance portfolio weight estimator based on the NLS is computed as

$$\hat{w}_{\mathrm{NLS}} = \frac{\widehat{\Sigma}_{\mathrm{NLS}}^{-1}\mathbf{1}}{\mathbf{1}^\top \widehat{\Sigma}_{\mathrm{NLS}}^{-1}\mathbf{1}}.$$

2.2. Sparse estimators

2.2.1. Dantzig selector

We consider a Dantzig-type approach (Candès and Tao, 2007) to estimate the matrix-vector product $\Sigma^{-1}\mathbf{1}_p$. Specifically, we aim to estimate the vector $\beta = \Sigma^{-1}\mathbf{1}_p$, which satisfies

$$\Sigma\beta = \mathbf{1},$$

under the assumption that $\beta \in \mathbb{R}^p$ is sparse. That is, we assume that only a small number of entries in β are nonzero.

To estimate β, we solve the following optimization problem:

$$\widehat{\beta} = \underset{\beta\in\mathbb{R}^p}{\operatorname{argmin}} \|\beta\|_1 \quad \text{subject to} \quad \|\mathbf{1} - S\beta\|_\infty \le \tau,$$

$\|\cdot\|_1$ denotes the ℓ_1-norm, $\|\cdot\|_\infty$ is the elementwise maximum norm, and $\tau > 0$ is a tuning parameter that controls the approximation tolerance.

This problem can be efficiently solved via linear programming. The resulting estimate $\widehat{\beta}$ serves as an approximation to $\Sigma^{-1}\mathbf{1}$. The minimum variance portfolio weight estimator is given by

$$\hat{w}_{\mathrm{dantzig}} = \frac{\hat{\beta}}{\mathbf{1}^\top \hat{\beta}}.$$

2.2.2. Graphical lasso

We consider the graphical lasso (glasso) method proposed by Friedman *et al.* (2008) for estimating a precision matrix. This method estimates the precision matrix $\Theta = \Sigma^{-1}$ by maximizing a penalized log-likelihood under the assumption that the matrix is sparse and the data are drawn from a multivariate Gaussian distribution. The penalization encourages sparsity in the off-diagonal elements of Θ, thereby reflecting conditional independence relationships between variables.

Formally, the estimator is obtained by solving the following optimization problem:

$$\widehat{\Theta} = \underset{\Theta \succ 0}{\operatorname{argmax}} \left\{ \log \det(\Theta) - \operatorname{tr}(S\,\Theta) - \lambda \|\Theta\|_1 \right\},$$

where S denotes the sample covariance matrix,

$$\|\Theta\|_1 = \sum_{i \neq j} |\theta_{ij}|$$

is the element-wise ℓ_1-norm excluding the diagonal, and $\lambda > 0$ is a regularization parameter that controls the level of sparsity.

Once $\widehat{\Theta}$ is obtained, it serves as an estimate of the inverse covariance matrix Σ^{-1}. The minimum variance portfolio weight estimator is given by

$$\hat{w}_{\text{glasso}} = \frac{\widehat{\Theta}\mathbf{1}}{\mathbf{1}^\top \widehat{\Theta}\mathbf{1}}$$

3. Exploration

3.1. Constructing a portfolio

To construct a portfolio, let $P_{i,t}$ denote the price of asset i at time t. The return of asset i at time t is computed as

$$x_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}}.$$

The return vector at time t, consisting of p assets, is denoted by

$$X_t = (x_{1,t}, \ldots, x_{p,t})^\top \in \mathbb{R}^p.$$

For each evaluation day T^*, portfolio weights are estimated using the most recent n observations. Specifically, we define the historical return matrix as

$$X^{(T^*)} = [X_{T^*-n}, \ldots, X_{T^*-1}] \in \mathbb{R}^{p \times n}.$$

Let $\mathcal{M}$ denote one of the estimation methods considered in Section 2, such as POET, JS, NLS, graphical lasso, or the Dantzig-type estimator. We define

$$\hat{w}^{(T^*)} = W\left(X^{(T^*)}; \mathcal{M}\right),$$

where $W(\cdot; \mathcal{M})$ denotes the full weight construction rule induced by method $\mathcal{M}$.

The exact meaning of $W(\cdot; \mathcal{M})$ depends on the object estimated by the method. For covariance-based methods, such as POET, JS, and NLS, the method first estimates a covariance matrix $\widehat{\Sigma}_{\mathcal{M}}$ from $X^{(T^*)}$, and the portfolio weight is computed as

$$\hat{w}^{(T^*)} = \frac{\widehat{\Sigma}_{\mathcal{M}}^{-1}\mathbf{1}}{\mathbf{1}^\top \widehat{\Sigma}_{\mathcal{M}}^{-1}\mathbf{1}}.$$

For graphical lasso, the method estimates a precision matrix $\widehat{\Theta}_{\mathcal{M}}$, and the corresponding weight is computed as

$$\hat{w}^{(T^*)} = \frac{\widehat{\Theta}_{\mathcal{M}}\mathbf{1}}{\mathbf{1}^\top \widehat{\Theta}_{\mathcal{M}}\mathbf{1}}.$$

For the Dantzig-type estimator, the method directly estimates the portfolio-relevant vector

$$\beta = \Sigma^{-1}\mathbf{1}.$$

If $\widehat{\beta}_{\mathcal{M}}$ denotes this estimate, the portfolio weight is computed by normalizing it as

$$\hat{w}^{(T^*)} = \frac{\widehat{\beta}_{\mathcal{M}}}{\mathbf{1}^\top \widehat{\beta}_{\mathcal{M}}}.$$

In all cases, the resulting weight vector satisfies the budget constraint

$$\mathbf{1}^\top \hat{w}^{(T^*)} = 1.$$

The realized portfolio return on day T^* is then obtained by applying the estimated weight to the out-of-sample return vector X_{T^*}:

$$r^{(T^*)} = \left(\hat{w}^{(T^*)}\right)^\top X_{T^*} \in \mathbb{R}.$$

Thus, the portfolio construction procedure can be summarized as follows:

1. $X^{(T^*)} = [X_{T^*-n}, \ldots, X_{T^*-1}] \in \mathbb{R}^{p\times n}$,
2. $\hat{w}^{(T^*)} = W\left(X^{(T^*)}; \mathcal{M}\right)$,
3. $r^{(T^*)} = \left(\hat{w}^{(T^*)}\right)^\top X_{T^*} \in \mathbb{R}$.

3.2. Data

The empirical analysis is based on two equity-index datasets, KOSPI and S&P 500. Daily adjusted closing prices were obtained using the `quantmod` package in R over the six-year period from January 1, 2019, to December 31, 2024, excluding the period from January 1, 2022, to December 31, 2022, due to inconsistencies in the collected price data.

In the preprocessing step, we first removed individual observations with missing price values. Daily returns were then calculated. We then excluded assets whose returns remained zero for more than 30 consecutive trading days. Such prolonged zero-return periods can lead to distorted covariance estimation and overly concentrated portfolio weights. The 30-day threshold was used to distinguish prolonged non-trading periods, such as periods before listing, after delisting, or extended trading suspensions, from short-term data irregularities or temporary suspensions which have only limited

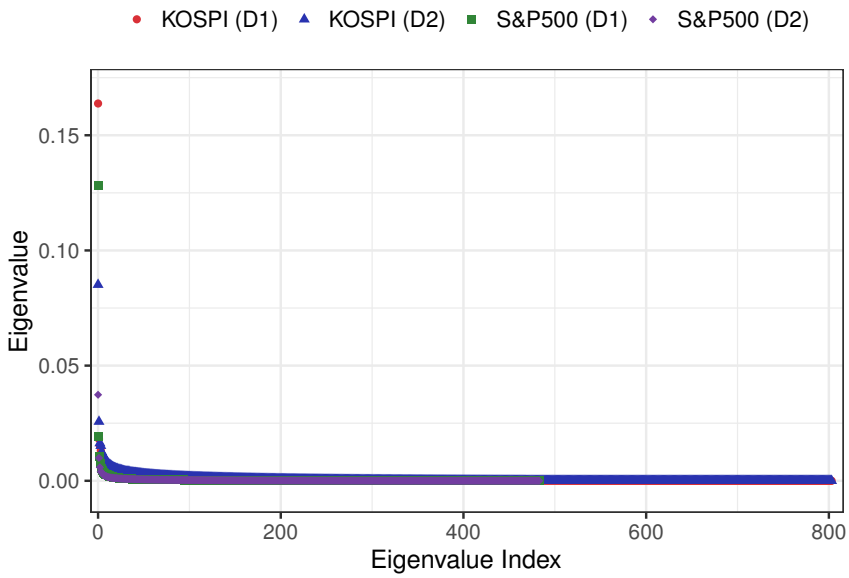


Figure 1: *Scree plots of sample covariance matrices for each dataset.*

effects on estimation. After this screening procedure, the final datasets contain 804 assets for KOSPI and 484 assets for S&P 500. This fixed set of assets provides a consistent data setting for comparing the methods, although it may not fully reflect a realistic investment setting where investable assets change over time.

To account for the effect of COVID-19, we divide the sample into two subperiods: a pandemic-period sample, spanning from January 1, 2019, to December 31, 2021, and a non-pandemic sample, spanning from January 1, 2023, to December 31, 2024. We refer to these samples as D1 and D2, respectively, for each of the KOSPI and S&P 500. We therefore analyze each market using two samples, one including the pandemic and the other excluding it. After the preprocessing and sample splitting, the KOSPI-D1 and KOSPI-D2 datasets contain 739 and 470 observations, respectively, with 804 assets in each dataset. The S&P 500-D1 and S&P 500-D2 datasets contain 756 and 501 observations, respectively, with 484 assets in each dataset.

Figure 1 presents scree plots of the sample covariance matrices for the four datasets. The plots show that both the number and magnitude of leading eigenvalues differ substantially across the two samples. In particular, the leading eigenvalue is larger when the pandemic period is included, indicating a stronger common component and increased market co-movement. This suggests that the underlying covariance structure depends on whether the pandemic period is included in the sample. It is worth examining whether the performance of covariance estimation methods also varies across the two sample designs.

3.3. Evaluating portfolio risk

Suppose that a dataset consists of p assets and N daily return observations, and that an estimation method $\mathcal{M}$ is fixed. For each evaluation day $T^* = n+1, \ldots, N$, portfolio weights are estimated using

the most recent n observations,

$$X^{(T^*)} = [X_{T^*-n}, \ldots, X_{T^*-1}] \in \mathbb{R}^{p\times n}.$$

The estimated weight vector is given by

$$\hat{w}^{(T^*)} = W\left(X^{(T^*)}; \mathcal{M}\right),$$

and the corresponding out-of-sample realized return is computed as

$$r^{(T^*)} = \left(\hat{w}^{(T^*)}\right)^\top X_{T^*}, \quad T^* = n+1, \ldots, N.$$

This rolling-window procedure produces a sequence of daily updated portfolio weights and corresponding out-of-sample realized returns. For each method and dataset, portfolio risk is evaluated by the empirical standard deviation of these realized returns:

$$\widehat{\sigma}_{\mathcal{M}} = \left\{\frac{1}{N-n-1}\sum_{T^*=n+1}^{N}\left(r^{(T^*)} - \bar{r}\right)^2\right\}^{1/2},$$

where

$$\bar{r} = \frac{1}{N-n}\sum_{T^*=n+1}^{N} r^{(T^*)}.$$

The empirical experiments compare the portfolio risk of each method across datasets. We consider two rolling-window lengths: $n = 100$, corresponding to approximately three months of daily returns, and $n = 250$, corresponding to approximately one year of daily returns.

3.4. Choice of parameters

We specify the reference direction used for JS and JS-diag in this experiment. We use a single reference direction,

$$v_1 = \frac{1}{\sqrt{p}}(1, \ldots, 1)^\top \in \mathbb{R}^p.$$

For methods based on a low rank plus sparse structure, such as POET and the JS, the number of leading components K must be specified. Although the number of spiked eigenvalues provides a natural reference, it does not necessarily minimize the out-of-sample portfolio risk. In practice, using a larger value of K may improve performance by capturing additional directions that are relevant for portfolio construction.

To explore this effect empirically, we varied the number of components K from 1 to 30 for both POET and JS, and computed the realized portfolio risk for each fixed value of K. Across all four datasets, the empirical standard deviation of realized returns generally decreased as K increased, while the marginal improvement became negligible beyond a certain point. This stabilization was consistently observed around $K = 14$ or $K = 15$. Based on this finding, we fixed $K = 15$ for both POET and JS in the subsequent analysis. We also adopt $K = 15$ for JS-diag to make the comparison with the JS. To check the robustness of the choice of $K = 15$, see Appendix D.

For the remaining tuning parameters, including those in the Dantzig-type estimator and graphical lasso, there is no closed-form choice known to minimize portfolio risk. We therefore considered a grid of candidate values and reported the best-performing result for each method and dataset. The grid-search results are reported in Appendices B and C.

Table 1: Experiment results for KOSPI data. All values in the table are multiplied by 10^3 for readability. Values outside the parentheses represent the standard deviation (*sd*) of calculated daily returns, and values inside parentheses are the standard error of *sd*.

Data	**KOSPI D1**				**KOSPI D2**			
Length	**250**		**100**		**250**		**100**	
Estimator	*sd*	(*se*)	*sd*	(*se*)	*sd*	(*se*)	*sd*	(*se*)
POET	5.329	(0.1416)	**4.992**	**(0.1394)**	3.439	(0.0595)	**3.918**	**(0.0938)**
JS	5.282	(0.1040)	5.278	(0.1003)	**3.254**	**(0.0265)**	3.925	(0.0484)
JS-diag	**5.236**	**(0.0896)**	114.225	(316.3562)	3.603	(0.0278)	27.719	(5.8869)
NLS	7.166	(0.2044)	7.277	(0.2027)	4.699	(0.0984)	5.910	(0.1694)
glasso	12.644	(0.4755)	11.203	(0.3681)	8.760	(0.2560)	8.717	(0.2697)
dantzig	15.884	(0.7349)	13.761	(0.6212)	10.501	(0.370)	11.659	(0.4283)

Table 2: Experiment results for S&P500 data. All values in the table are multiplied by 10^3 for readability. Values outside the parentheses represent the standard deviation (*sd*) of calculated daily returns, and values inside parentheses are the standard error of *sd*.

Data	**S&P500 D1**				**S&P500 D2**			
Length	**250**		**100**		**250**		**100**	
Estimator	*sd*	(*se*)	*sd*	(*se*)	*sd*	(*se*)	*sd*	(*se*)
POET	**10.757**	**(0.4409)**	**9.251**	**(0.3243)**	**5.133**	**(0.0346)**	**5.168**	**(0.4530)**
JS	10.768	(0.3963)	9.387	(0.2934)	5.244	(0.0356)	5.753	(0.0577)
JS-diag	11.356	(0.5474)	13.059	(1.1281)	5.537	(0.0413)	6.558	(0.0829)
NLS	10.871	(0.4363)	9.735	(0.3542)	5.309	(0.0451)	5.410	(0.0527)
glasso	13.331	(0.5683)	11.458	(0.4367)	6.153	(0.0668)	6.399	(0.0634)
dantzig	15.614	(0.6557)	13.813	(0.7083)	7.105	(0.0879)	7.123	(0.0801)

3.5. Result

Table 1 and Table 2 report the empirical standard deviation of out-of-sample realized returns for each estimator. All reported values are multiplied by 10^3 for readability. For each method and dataset, the table reports the best result over the tuning parameter values considered.

The first notable pattern is that dense covariance-based estimators consistently outperform sparsity-inducing methods. Across both the KOSPI and S&P 500 datasets, POET, JS, JS-diag, and NLS generally produce lower realized standard deviations than graphical lasso and the Dantzig-type estimator. This suggests that imposing sparsity directly on the precision matrix or on the portfolio-relevant vector $\Sigma^{-1}\mathbf{1}$ may lead to overly concentrated portfolio allocations. In contrast, dense covariance-based estimators preserve broader diversification, which appears to be more effective for reducing out-of-sample risk in the minimum variance portfolio problem considered here.

Among the dense methods, estimators that explicitly incorporate the leading components of the covariance matrix perform particularly well. POET is the most robust method overall, attaining the lowest realized standard deviation in all S&P 500 cases and in most of the KOSPI cases. The JS-based estimator also performs competitively, and it achieves the lowest standard deviation in some KOSPI settings. This indicates that the leading common component of asset returns plays an important role in portfolio construction. Methods that combine a low-rank leading component with regularization of the residual covariance structure are therefore well suited to the datasets considered in this study.

The comparison between JS and JS-diag further highlights the importance of how the residual covariance component is treated. Although JS-diag performs well in some cases, it becomes highly unstable for the KOSPI dataset with $n = 100$, producing much larger standard deviations than the other dense methods. This suggests that replacing the residual covariance matrix by a purely diagonal

structure may be too restrictive, especially when the estimation window is short. Allowing for a regularized residual dependence structure, as in POET and the JS estimator with thresholding, appears to produce more stable portfolio weights. In fact, the resulting portfolio weight from JS-diag with short rolling window can be overly concentrated, which is in part due to numerical instability with smaller sample sizes. For further analysis on weights of JS and JS-diag on KOSPI with short rolling window, see Appendix F.

The effect of the rolling-window length also depends on whether the pandemic period is included. For the pandemic-period samples, the shorter window $n = 100$ generally produces lower realized standard deviation than $n = 250$. This is consistent with the view that the period of intense market-wide fluctuation was relatively short, so a longer window may mix observations from different market conditions and lead to less stable weight estimates. In contrast, for the non-pandemic samples, the longer window $n = 250$ is generally more favorable, presumably because the covariance structure is more stable and can be estimated more accurately from a larger number of observations.

Overall, the results support two main conclusions. First, dense portfolio allocations outperform sparse allocations in these high-dimensional minimum variance portfolio experiments. Second, covariance estimators that preserve and regularize the leading common component, while also controlling the residual covariance structure, provide the most stable out-of-sample portfolio performance. As a supplementary return-based performance measure, annualized Sharpe ratios are also reported in Appendix E.

4. Discussion

A natural extension of the global minimum variance portfolio is obtained by imposing an additional expected return constraint. Although this study focuses on the global minimum variance portfolio with only the fully invested constraint, the covariance estimation framework considered here can be extended to the mean-variance portfolio problem. Let $\mu \in \mathbb{R}^p$ denotes the vector of expected asset returns and let μ_0 be the target portfolio return. The corresponding optimization problem is

$$w^* = \underset{w \in \mathbb{R}^p}{\operatorname{argmin}}\, w^\top \Sigma w \quad \text{subject to} \quad w^\top \mathbf{1} = 1, \quad w^\top \mu = \mu_0.$$

Define

$$A = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}, \quad B = \mathbf{1}^\top \Sigma^{-1} \mu, \quad C = \mu^\top \Sigma^{-1} \mu, \quad D = AC - B^2.$$

Then the closed form solution is given by

$$w^* = \frac{C - B\mu_0}{D} \Sigma^{-1} \mathbf{1} + \frac{A\mu_0 - B}{D} \Sigma^{-1} \mu.$$

This expression shows that portfolio weight still depends on the inverse covariance matrix. Therefore, once a covariance matrix estimator is obtained by any of the methods considered in this study, it can be plugged into the above formula in place of Σ. In this sense, the empirical comparison conducted in this study provides a basis for extending the same covariance estimation methods to portfolio optimization problems with additional constraints.

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Appendix A: Implementation details for NLS

Using Epanechnikov kernel, the estimated non-linear shrinkage function for sample eigenvalues is as follows:

$$\hat{d}(x) = \frac{x}{\left[\pi \frac{p}{n} x \hat{f}(x)\right]^2 + \left[1 - \frac{p}{n} - \pi \frac{p}{n} x \mathcal{H}_{\hat{f}}(x)\right]^2},$$

where,

$$\hat{f}(\hat{\lambda}_i) := \frac{1}{p} \sum_{j=1}^{p} \frac{3}{4\sqrt{5}\, h_j} \left[1 - \frac{1}{5} \left(\frac{\hat{\lambda}_i - \hat{\lambda}_j}{h_j} \right)^2 \right]_+, \quad i = 1, \ldots, p,$$

$$\mathcal{H}_{\hat{f}}(\hat{\lambda}_i) := \frac{1}{p} \sum_{j=1}^{p} \left\{ -\frac{3(\hat{\lambda}_i - \hat{\lambda}_j)}{10\pi h_j^2} + \frac{3}{4\sqrt{5}\pi h_j} \left[1 - \tfrac{1}{5} \left(\tfrac{\hat{\lambda}_i - \hat{\lambda}_j}{h_j} \right)^2 \right] \right\}$$
$$\times \log \left| \frac{\sqrt{5}\, h_j - \hat{\lambda}_i + \hat{\lambda}_j}{\sqrt{5}\, h_j + \hat{\lambda}_i - \hat{\lambda}_j} \right|.$$

We set $h_j = T^{-1/3} \hat{\lambda}_j, \quad j = 1, \ldots, p$, as in the default implementation of the nonlinear shrinkage estimator (Ledoit and Wolf, 2020). This choice makes the bandwidth proportional to the local scale of $\hat{\lambda}_j$ and no additional bandwidth tuning is performed.

Appendix B: Grid search for the Dantzig-type estimator

For the Dantzig-type estimator, we used the common tuning grid $\tau \in \{0.05, 0.10, 0.20, 0.50, 0.80\}$ for all datasets and rolling-window lengths.

Table B.1: Tuning parameter analysis for the Dantzig-type estimator on the KOSPI data. All reported empirical standard deviations are multiplied by 10^3 for readability

Data	**KOSPI D1**		**KOSPI D2**	
Length	**250**	**100**	**250**	**100**
Estimator	*sd*		*sd*	
$\tau = 0.05$	16.734	14.312	12.510	12.002
$\tau = 0.1$	16.801	14.620	12.381	12.468
$\tau = 0.2$	**15.884**	**13.761**	12.017	11.804
$\tau = 0.5$	17.331	14.189	**10.501**	**11.659**
$\tau = 0.8$	19.644	15.096	11.398	12.170

Table B.2: Tuning parameter analysis for the Dantzig-type estimator on the S&P500 data. All reported empirical standard deviations are multiplied by 10^3 for readability.

Data	**S&P500 D1**		**S&P500 D2**	
Length	**250**	**100**	**250**	**100**
Estimator	*sd*		*sd*	
$\tau = 0.05$	16.924	15.678	7.054	**7.123**
$\tau = 0.1$	15.814	15.945	7.494	7.231
$\tau = 0.2$	**15.614**	**13.813**	7.281	7.278
$\tau = 0.5$	18.480	15.202	**7.105**	7.435
$\tau = 0.8$	18.274	16.060	7.246	7.661

Appendix C: Grid search for graphical lasso

For graphical lasso, the tuning grid was constructed using the `huge` package. For each dataset, `huge` computes a data-specific $\lambda_{\max}$, defined as the smallest value of λ that yields an empty graph, and we used 10 candidate values equally spaced on the logarithmic scale from $\lambda_{\max}$ to $0.1\lambda_{\max}$. This grid was computed once for each dataset and was used commonly across all rolling windows, rather than being recalculated for each rolling window. Since $\lambda_{\max}$ depends on the empirical covariance structure, the graphical lasso tuning grid differs across datasets.

Table C.1: Tuning parameter analysis for graphical lasso on the KOSPI data. All reported empirical standard deviations are multiplied by 10^3 for readability.

Data	**KOSPI D1**			**KOSPI D2**	
Length	**250**	**100**		**250**	**100**
Estimator	*sd*			*sd*	
$\lambda = 0.905$	15.599	14.333	$\lambda = 0.948$	10.405	10.051
$\lambda = 0.701$	15.510	13.946	$\lambda = 0.734$	10.401	10.026
$\lambda = 0.543$	15.154	13.374	$\lambda = 0.568$	10.353	9.828
$\lambda = 0.420$	14.724	12.730	$\lambda = 0.440$	10.186	9.497
$\lambda = 0.325$	14.262	12.125	$\lambda = 0.341$	9.904	9.053
$\lambda = 0.252$	13.804	11.695	$\lambda = 0.264$	9.408	**8.717**
$\lambda = 0.195$	13.459	11.425	$\lambda = 0.204$	8.975	8.772
$\lambda = 0.151$	13.161	11.239	$\lambda = 0.158$	**8.760**	9.005
$\lambda = 0.117$	12.817	**11.203**	$\lambda = 0.122$	8.796	9.101
$\lambda = 0.091$	**12.644**	11.240	$\lambda = 0.095$	8.925	9.144

Table C.2: Tuning parameter analysis for graphical lasso on the S&P 500 data. All reported empirical standard deviations are multiplied by 10^3 for readability

Data	**S&P D1**			**S&P D2**	
Length	**250**	**100**		**250**	**100**
Estimator	*sd*			*sd*	
$\lambda = 0.993$	17.923	16.166	$\lambda = 0.998$	7.259	7.506
$\lambda = 0.769$	17.472	15.335	$\lambda = 0.772$	7.255	7.490
$\lambda = 0.595$	15.277	13.194	$\lambda = 0.598$	7.216	7.278
$\lambda = 0.461$	14.038	12.074	$\lambda = 0.463$	6.878	6.758
$\lambda = 0.357$	13.475	11.483	$\lambda = 0.359$	6.397	6.405
$\lambda = 0.276$	**13.331**	**11.458**	$\lambda = 0.278$	**6.153**	**6.399**
$\lambda = 0.214$	13.535	11.913	$\lambda = 0.215$	6.201	6.624
$\lambda = 0.166$	14.023	12.570	$\lambda = 0.166$	6.511	6.882
$\lambda = 0.128$	14.742	13.153	$\lambda = 0.129$	6.920	7.012
$\lambda = 0.099$	15.353	13.450	$\lambda = 0.100$	7.229	7.074

Appendix D: Robustness check for the choice of the number of principal components

To assess the robustness of the number of principal components $K = 15$ in POET and JS method, we report the results of nearby values

$$K \in \{11, 13, 15, 17, 19\}.$$

Table D.1, Table D.2 show that the empirical standard deviations of POET and the JS-based estimator remain relatively stable for values of K near 15, although the numerically best value varies slightly

across datasets and rolling-window lengths. This robustness check supports the use of $K = 15$ as a representative choice in the main analysis.

Table D.1: Robustness check for the choice of K in POET and the JS-based estimator on the KOSPI data. All reported empirical standard deviations are multiplied by 10^3 for readability.

Data	**KOSPI D1**				**KOSPI D2**			
Length	250		100		250		100	
Method	POET	JS	POET	JS	POET	JS	POET	JS
Estimator	*sd*		*sd*		*sd*		*sd*	
$K = 11$	5.459	5.296	5.036	5.386	3.464	3.218	3.846	3.897
$K = 13$	5.416	5.226	5.013	5.329	3.456	3.192	3.9	3.905
$K = 15$	5.329	5.282	4.992	5.278	3.439	3.254	3.918	3.925
$K = 17$	5.426	5.258	4.992	5.330	3.45	3.172	3.937	3.903
$K = 19$	5.424	5.280	5.03	5.321	3.408	3.125	3.955	3.930

Table D.2: Robustness check for the choice of K in POET and the JS-based estimator on the S&P 500 data. All reported empirical standard deviations are multiplied by 10^3 for readability.

Data	**S&P500 D1**				**S&P500 D2**			
Length	250		100		250		100	
Method	POET	JS	POET	JS	POET	JS	POET	JS
Estimator	*sd*		*sd*		*sd*		*sd*	
$K = 11$	11.674	10.649	9.934	9.496	5.143	5.251	5.374	6.070
$K = 13$	10.672	10.371	9.499	9.417	5.119	5.155	5.299	5.958
$K = 15$	10.757	10.768	9.251	9.387	5.133	5.244	5.168	5.753
$K = 17$	10.680	10.542	9.370	9.482	5.051	5.149	5.289	5.798
$K = 19$	10.499	10.458	9.334	9.551	5.042	5.185	5.282	5.866

Appendix E: Analysis of annualized Sharpe ratios

In addition to portfolio risk, we report annualized Sharpe ratios as supplementary return-based performance measures. The annualized Sharpe ratio is computed as

$$\mathrm{SR}_{\mathrm{ann}} = \frac{\bar{r}}{\hat{\sigma}} \times \sqrt{252},$$

where $\bar{r}$ is the average daily realized return and $\hat{\sigma}$ is the empirical standard deviation of daily realized returns. In this analysis, the risk-free rate is set to zero.

Table E.1: Annualized Sharpe ratios for the KOSPI data

Data	**KOSPI D1**		**KOSPI D2**	
Length	250	100	250	100
Estimator	$\mathrm{SR}_{\mathrm{ann}}$		$\mathrm{SR}_{\mathrm{ann}}$	
POET	0.758	0.003	1.239	-0.618
JS	0.618	-0.068	1.831	-0.440
JS-diag	0.558	-0.485	1.956	0.145
NLS	0.398	0.257	0.227	-0.643
glasso	0.202	0.007	-0.318	-1.081
dantzig	0.780	0.949	0.516	-0.272

Table E.2: Annualized Sharpe ratios for the S&P500 data

Data	**S&P500 D1**		**S&P500 D2**	
Length	250	100	250	100
Estimator	SR_{ann}		SR_{ann}	
POET	0.662	1.085	2.015	1.180
JS	0.359	1.232	2.143	1.461
JS-diag	0.714	0.698	2.034	0.862
NLS	0.432	1.200	2.941	1.319
glasso	1.633	1.964	1.620	1.309
dantzig	1.765	1.548	2.732	1.758

The Sharpe ratio results are reported as supplementary return-based performance measures. Because the Sharpe ratio depends on both average realized return and realized volatility, the rankings are not necessarily identical to those based on empirical standard deviation.

Appendix F: Weight comparison for JS and JS-diag

In the main analysis (Table 1), JS-diag showed unstable performance for the KOSPI data with the short rolling window ($n = 100$). To further examine whether this instability is related to portfolio concentration, we compare the weights obtained from JS-diag with those obtained from JS for KOSPI-D1 and KOSPI-D2 when $n = 100$.

Let $w_t = (w_{1t}, \ldots, w_{pt})^\top$ denote the portfolio weight vector obtained on evaluation day $t = 1, \ldots, T$ for p assets. We summarize the concentration of w_t using three quantities: the proportion of numerically nonzero entries, the maximum absolute entry, and the ℓ_1-norm,

$$
\begin{aligned}
p_{\epsilon,t} &= \frac{1}{p} \sum_{i=1}^{p} \mathbf{1}\{|w_{it}| > \epsilon\}, \quad \epsilon = 10^{-3}, \\
m_t &= \max_{1 \le i \le p} |w_{it}|, \\
\|w_t\|_1 &= \sum_{i=1}^{p} |w_{it}|.
\end{aligned}
$$

We report the averages of these quantities over the evaluation period. Specifically, we define $\bar{p}_\epsilon = \frac{1}{T} \sum_{t=1}^{T} p_{\epsilon,t}$, $\bar{m} = \frac{1}{T} \sum_{t=1}^{T} m_t$, and $L_1 = \frac{1}{T} \sum_{t=1}^{T} \|w_t\|_1$.

Table F.1: Weight concentration diagnostics for JS and JS-diag on the KOSPI data with the short rolling window ($n = 100$).

Data	**KOSPI D1**		**KOSPI D2**	
Length	100		100	
Method	JS	JS-diag	JS	JS-diag
$\bar{p}_\epsilon$	0.7326	0.6527	0.6967	0.6523
$\bar{m}$	0.0749	2.0833	0.0556	1.7834
L_1	3.0529	7.6600	2.8460	7.8022

Table F.1 indicates that JS-diag tends to produce more concentrated portfolio weights than JS in both KOSPI-D1 and KOSPI-D2. In particular, JS-diag has much larger values of $\bar{m}$ and L_1, suggesting that some individual weights become large in absolute value. This provides a possible explanation for the unstable out-of-sample performance of JS-diag in the short-window KOSPI setting.

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